

muchísimocks: An emulated cosmological simulation suite for simulation-based inference

Kate Storey-Fisher,^{1,2*} Raúl Angulo,^{2,3} Marcos Pellejero-Ibañez,⁴ and friends [ksf: auth order?](#)

¹*Kavli Institute for Particle Astrophysics and Cosmology, Stanford University, 452 Lomita Mall, Stanford, CA 94305, USA*

²*Donostia International Physics Center, Paseo Manuel de Lardizabal 4, 20018 Donostia-San Sebastian, Spain*

³*IKERBASQUE, Basque Foundation for Science, E-48013, Bilbao, Spain*

⁴*Institute for Astronomy, University of Edinburgh, Royal Observatory, Blackford Hill, Edinburgh, EH9 3HJ, UK*

Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

Galaxy clustering contains rich cosmological information down to non-linear scales; however, methods to extract this information are currently limited by the high cost of cosmological N -body simulations, as well as by the fidelity of galaxy bias modeling. We use a new approach to generating approximate but robust galaxy mocks: we employ the `map2map` full-field emulator to generate particle displacement and velocity fields, and apply the hybrid Lagrangian bias expansion approach to flexibly model biased tracers. With this forward model, we create a suite of mock 3D galaxy density fields each of size $(1\ h^{-1}\ \text{Gpc})^3$ at $z = 0$ spanning 10,000 different cosmologies, called **muchísimocks**; we construct tracer fields with up to 32 bias parameters for cosmology, resulting in a 320,000-mock training set. We train a simulation-based inference (SBI) model on summary statistics of the density fields—the galaxy–galaxy power spectrum, the galaxy–matter power spectrum, and the galaxy bispectrum—and infer the cosmological parameters, marginalizing over the bias and noise coefficients. We validate the pipeline on in-distribution test mocks and find our inferred posteriors to be well-calibrated across the prior range. With this large suite, we are able to test convergence with the number of simulations: we find that while 20,000 training points are sufficient for the power spectrum, when including the additional statistics precision continues to improve beyond 320,000, and that increasing the number of bias parameter sets per cosmology yields similar constraints as increasing the number of cosmologies. We also show that we recover unbiased cosmological constraints from an out-of-distribution model, an independent high-resolution N -body simulation populated with galaxies using extended SubHalo Abundance Matching, demonstrating the generalizability of the approach. We find that the galaxy–matter power spectrum is critical for breaking parameter degeneracies, and additionally including the bispectrum significantly improves precision, on σ_8 by XXX and b_1 by XXX. These results are useful for guiding the design of future SBI approaches for galaxy clustering, and demonstrate a promising approach for full-field inference as well as a range of other potential applications. The **muchísimocks** suite and the code used in this paper are available at [thisurl](#).

Key words: methods: numerical – methods: statistical – galaxies: statistics – cosmology: large-scale structure of Universe – cosmological parameters

1 INTRODUCTION

The clustering of galaxies encodes information about the cosmological model describing the universe. Surveys such as the Sloan Digital Sky Survey ? and most recently the Dark Energy Spectroscopic Instrument ? have spectroscopically observed millions of galaxies, producing detailed maps of a large volume of the observable universe. Current and upcoming surveys such as Euclid (?) and the Roman Observatory ? will deliver more and more high quality galaxy clustering data sets.

Standard approaches to analyzing this spectroscopic survey data rely on analytic techniques to model summary statistics of the galaxy field, and Monte Carlo techniques to sample the parameters of interest ??. Analyses are thus largely restricted to two-point statistics

on quasi-linear scales, which contain robust cosmological information but do not fully capture the information content in the data. For instance, the DESI analyses use a perturbation theory approach, and their analysis is limited to $k_{\text{max}} = \text{XXX}$ for the power spectrum and $k_{\text{max}} = \text{XXX}$ for the bispectrum. Further, as surveys become more sensitive and statistical uncertainties improve, systematic errors will become increasingly relevant, and the effect of various systematics on clustering statistics—both two-point and higher-order statistics—is difficult to model.

A new approach known as simulation-based inference (SBI), or likelihood-free inference, has shown promise for cosmological inference from galaxy clustering, as it obviates these limitations. In the SBI approach, only a forward model of the data is required, and the posterior of the input parameters is approximated conditioned on the observed data. In the case of galaxy clustering, one can thus forward model the full galaxy field and directly measure any statistic, without

* E-mail: kstoreyf@stanford.edu

the need for analytic modeling. Similarly, one can directly forward model the effects of systematics on the galaxy field, and the statistics simply measured without the need for additional modeling specific to each statistic. Finally, Monte Carlo methods typically assume a Gaussian form for the likelihood (and must estimate a nontrivial covariance matrix), even though higher-order statistics in particular are known to have non-Gaussian properties (Hahn et al. 2018); SBI makes no assumptions about the form of the likelihood. Further, in the case of galaxy clustering one can learn optimized statistics of the galaxy field for the parameters of interest, effectively using the information in the full galaxy field.

While promising, SBI faces challenges for galaxy clustering analyses. One key issue is that SBI typically requires many simulations, but cosmological simulations are very computationally expensive. An N -body simulation with the volume and resolution to accurately describe galaxy clustering into the nonlinear regime costs at least hundreds (and up to millions) of CPU hours, and thousands to hundreds of thousands of simulations are generally needed for SBI. Another challenge is modeling how galaxies populate the dark matter distribution; models must be relatively fast, flexible enough to model various galaxy populations, and accurate down to small scales.

There has been significant recent work that has demonstrated SBI as a promising path forward for the field. The SimBIG collaboration employs a forward model based on the $2,000 (1h^{-1} \text{ Gpc})^3$ simulations of the Quijote Latin hypercube (?); they use a halo occupation distribution (HOD, ?) model to populate the simulations with galaxies, choosing 10 HODs per cosmology for a total of 20,000 simulations (Hahn et al. 2023c). Using the power spectrum and applying to SDSS BOSS data (?), they achieved constraints on σ_8 27% more precise than the standard analysis thanks to the information in small scales (Hahn et al. 2022); subsequent analyses have applied SimBIG to the bispectrum (Hahn et al. 2023a), the marked power spectrum (Masara et al. 2024), and the full galaxy field (Lemos et al. 2023; Hahn et al. 2023b). Tucci & Schmidt (2024) generate 10^5 simulations based on the effective field theory of large-scale structure (EFTofLSS), and employ neural posterior estimation using the power spectrum and bispectrum out to $k_{\text{max}} = 0.2$; as this model is fully perturbative, they are able to use a Euclid-size volume of $63 (h^{-1} \text{ Gpc})^3$. By performing a direct comparison to standard inference, they find that the off-diagonal elements of the covariance matrix are important for constraining σ_8 accurately, while allowing for a non-Gaussian likelihood is less important. Using the same forward model, Nguyen et al. (2024) show that full-field inference yields over a factor of five improvement on the constraint on σ_8 compared to the power spectrum and bispectrum at the same scale cut of $k_{\text{max}} = 0.12$. Recent work has also explored forward models based on more complicated and physical models of galaxy formation, albeit at the expense of much smaller volumes. de Santi et al. (2023) employ the 1000-simulation Latin hypercubes of the CAMELS suite (?), which are hydrodynamical simulations with $(25h^{-1} \text{ Mpc})^3$ volumes, to perform SBI using graph neural networks; they find that their inference on Ω_m is robust to galaxy formation physics.

Given the high computational cost of cosmological simulations are, work has turned to developing emulators that learn the mapping from cosmological parameters to the results of the N -body simulation. Much work has been done on emulators of statistics of the density field, largely for the power spectrum. Recently, full-field emulators have been developed that output the particle displacement field. The emulator map2map (Jamieson et al. 2023) uses a convolutional neural network-based architecture to map from Zel’dovich approximation displacements to N -body displacements, trained on the Quijote simulation suite ?. It has been used in a range of appli-

cations, such as producing large-volume emulated simulations (?); variants of map2map have been designed to emulate modified gravity (?) and to learn the mapping from non-linear structures to the initial conditions of the universe (Doerer et al. 2024). There has been continued work on developing full-field emulators, with a range of training data and neural architectures (??).

Robustly the galaxy–dark matter connection is a key challenge in cosmological inference. Halo-based models such as the halo occupation distribution (?) populate dark matter halos with galaxies based on halo mass and properties, but their flexibility degrades their robustness, particularly for higher-order statistics (e.g. ??). More physically motivated models like subhalo-abundance matching (??) are more robust, but require merger trees which are more computationally expensive; even more complicated models like semi-analytic models (e.g. ??) and hydrodynamical simulations (e.g. ??) are both expensive and make specific choices regarding galaxy formation, which makes robustness a challenge. EFTofLSS offers a perturbative approach that is robust on large scales, but breaks down on small scales (?).

The hybrid Lagrangian bias expansion (or hybrid effective field theory) offers an approach that is accurate into the small-scale regime while remaining computationally relatively low-cost. It combines the Lagrangian bias expansion, which models the galaxy field as a summation of operators of the dark matter field, with particle displacement and velocity fields from N -body simulations to advect them to Eulerian space. This approach has been shown to be effective down to $k \sim 0.6$ (?Pellejero-Ibanez et al. 2022), and has been used to construct accurate emulators of the power spectrum (??Pellejero Ibanez et al. 2023). Pellejero Ibanez et al. (2023) showed that emulated N -body displacements with map2map can be used as inputs to the hybrid bias expansion to provide an accurate model of galaxy abundance. They found 1-2% precision on the power spectrum $k \sim 0.6$; while there were some biases at intermediate scales, these were absorbed by nuisance parameters describing stochasticity. Notably, they found that the robustness held with variations in the cosmological parameters, demonstrating promise for using this approach for cosmological inference.

[how many sims paper] Based on their ..., Tucci & Schmidt (2024) estimate that of order 10^5 simulations are required ...

In this paper, we construct a suite of 3-dimensional mock galaxy fields using our forward model based on the hybrid Lagrangian bias expansion with emulated displacements from the full-field map2map emulator. The suite, called muchisimocks, consists of both real- and redshift-space mocks at $z = 0$ and spans galaxy bias and cosmological parameter space. This suite opens the door to simulation-based inference for cosmology with a hybrid bias expansion-based galaxy field, as well as with a very large number of simulations given the speed of the emulator. We demonstrate this approach by training a neural posterior inference model on summary statistics computed on muchisimocks in real space; in this work we focus on the galaxy–galaxy power spectrum, the galaxy–matter cross power spectrum, and the galaxy bispectrum. Our maximal training set consists of 10,000 cosmologies and 32 bias parameter sets per cosmology, for a total of 320,000 training mocks; we investigate the dependence of the inference performance on the number of training mocks, with an eye towards guiding the development of future SBI pipelines.

This paper is organized as follows. In §2, we present our forward model for galaxy fields, including details of the N -body emulator, hybrid bias expansion, and noise model. We introduce the methodology used for our simulation-based inference pipeline in §3, including the summary statistics analyzed and the construction of our training set and various test sets. In §4 we show our results on these test sets,

as well as analyses of the scale dependence and convergence with size of training data. We discuss the implications of our results in §5, and summarize our findings and conclude in §6.

2 THE MUCHISIMOCKS SUITE

2.1 Emulating the displacement field with map2map

The nonlinear matter distribution can be simulated to high accuracy with N -body simulations. However, simulations with the resolution and volume required to analyze current galaxy surveys are very expensive, and training simulation-based inference models requires a very large of simulations. We thus turn to a field-level emulator of the matter distribution, map2map Jamieson et al. (2023).

The density field can be approximated at linear order by the Zel’dovich approximation (ZA, also known as first-order Lagrangian perturbation theory, 1LPT), based on a realization of an initial conditions field. The map2map emulator learns a mapping from the ZA field at $z = 0$ to the non-linear N -body displacement field at that redshift, effectively learning the correction from linear to non-linear displacements. The emulator is trained on 2000 simulations of the Quijote simulation suite that span a latin hypercube (LH) of five cosmological parameters: Ω_c , σ_8 , h , Ω_b , and n_s (?). Each has a volume of $(1 h^{-1}\text{Gpc})^3$ with 512^3 particles, giving particle mass resolutions in the range $2 \times 10^{11} - 1 \times 10^{12} h^{-1} M_\odot$. map2map also trains a separate network to predict the nonlinear velocity field from the ZA velocities. Once trained, map2map can generate the displacement and velocity fields for any input ZA field, taking a few minutes on a GPU for a $(1 h^{-1}\text{Gpc})^3$ volume. While we do not utilize the velocity fields in this work, as we limit to real-space analysis, these will be important for future analyses including redshift-space distortions as well as other applications involving the velocity distribution; we release them as part of the muchisimocks suite. [ksf: explain special velocity field weights used, for zspace](#)

In more detail, map2map consists of two convolutional neural networks (CNNs), one to predict the matter density field and the other to predict the velocity field. The particle displacements are structured on a 3D grid as they are functions of their initial Lagrangian space position, so the CNN processes the inputs and outputs as 3-channel images. As most of the cosmological parameters only affect the input power spectrum that is used to create the ZA field, these are not needed as additional inputs to the model. However, the matter density Ω_m affects the N -body evolution, so is included as an input “style” parameter; this is incorporated via the approach of the StyleGAN2 (?), in which the style parameters modulate the convolution kernels. The model uses a V-net-like architecture, with 3 layers of down-sampling and up-sampling, and leaky ReLU activations. For its loss function, it combines the log mean squared error of the Lagrangian particle displacements and Eulerian density field. We note that there is a newer version of map2map that includes redshift dependence, and predicts the velocity field as a derivative of the position field (Jamieson et al. 2024); however, as we focus on $z = 0$ and also require a modification of the velocity field as described above, we use the original map2map implementation.

For our mock suite, we define a latin hypercube of 10,000 sets of cosmological parameters: Ω_c , σ_8 , h , Ω_b , and n_s . We use uniform priors for each parameter that span ~ 10 around the ? values of the cosmological parameters, following ? (with a range for σ_8 extended to slightly lower values); these are shown in Table 1. For each of these, we generate a ZA field based on a different random seed, and pass that field through map2map to obtain the nonlinear particle

displacements. The map2map-emulated particle displacements are a key ingredient in modeling the galaxy bias fields, as discussed in the next section.

2.2 Modeling tracers with the hybrid bias expansion

To model biased tracers of the matter field, we use the hybrid Lagrangian bias expansion, also known as hybrid effective field theory (HEFT, ???). We first define a mapping from initial (Lagrangian) space coordinates $\mathbf{q}$ to Eulerian space coordinates $\mathbf{x}$ as $\mathbf{x} = \mathbf{q} + \Psi(\mathbf{q})$, where $\Psi(\mathbf{q})$ is the Lagrangian space displacement field. This advection from Lagrangian to Eulerian space effectively expresses the overdensity field as weighted contributions from the initial density field that end up in a given Eulerian location. The displacement term $\Psi(\mathbf{q})$ can be computed with Lagrangian perturbation theory, but this is only accurate down to mildly non-linear scales; here we take the “hybrid” approach which uses N -body displacements to compute $\Psi(\mathbf{q})$, significantly improving the accuracy down to non-linear scales.

The galaxy overdensity field in Lagrangian space can be described to second order as

$$\delta_g(\mathbf{q}) = 1 + b_1 \delta_L(\mathbf{q}) + b_2 \left[\delta_L^2(\mathbf{q}) - \langle \delta_L^2 \rangle \right] + b_{s^2} \left[s^2(\mathbf{q}) - \langle s^2 \rangle \right] + b_\Delta \nabla^2 \delta_L(\mathbf{q}) \quad (1)$$

where $\delta_L(\mathbf{q})$ is the linear matter density in Lagrangian space, and $s^2(\mathbf{q})$ is the shear field, defined by the traceless part of the tidal field, given by $s^2(\mathbf{q}) = s_{ij}(\mathbf{q}) s^{ij}(\mathbf{q}) = \delta_i \delta_j \phi(\mathbf{q}) - \frac{1}{3} \delta_{ij}^K \delta_L(\mathbf{q})$, where $\phi(\mathbf{q})$ is the linear gravitational potential in Lagrangian space $\nabla^2 \phi(\mathbf{q}) = 4\pi G \bar{\rho} \delta_L(\mathbf{q})$, and δ^K is the Kronecker delta. The coefficients b_1 , b_2 , b_{s^2} , and b_Δ are the Lagrangian bias parameters. The prior ranges of the bias parameters used in our SBI analysis are shown in Table 1, though we note that a tracer field could be constructed for any choice of bias parameters with our saved bias fields.

We build this field starting from a random realization of the initial Gaussian field on a $N_{\text{grid}} = 512^3$ grid in a $L = (1 h^{-1} \text{Gpc})^3$ box, matching the resolution and size of the Quijote simulations that map2map was trained on, and use the ZA to compute approximate displacements. We pass the ZA displacement and velocity fields as input to map2map (§2.1), which outputs emulated N -body displacement and velocity fields. We compute the Lagrangian operators based on the linear density field, smoothing these with a damping scale of $k = 0.75 h \text{Mpc}^{-1}$. We then perform the advection to Eulerian space: The tracers are deposited onto a grid in Eulerian space with a cloud-in-cell (CIC) assignment, with positions given by the simulation displacement field and weights given by Lagrangian operator. This gives five bias fields in Eulerian space, O_i : weighting the Lagrangian density field by the constant 1 gives the homogeneous field; by $\delta_L(\mathbf{q})$, the linear density field; by $\delta_L^2(\mathbf{q}) - \langle \delta^2 \rangle$, the quadratic density field; by $s^2(\mathbf{q}) - \langle s^2 \rangle$, the tidal field; and by $\Delta^2 \delta(\mathbf{q})$, the Laplacian. These fields can be weighted by the bias coefficients and summed to produce an overall tracer field δ_g .

We perform two postprocessing steps on these fields. [ksf: fill in](#) To reduce storage space, we reduce the size of these fields by cutting the high- k modes: we remove modes beyond $k_{\text{Nyquist}} = \pi N_{\text{grid, cut}} / L \approx 0.4$, where $N_{\text{grid, cut}} = 128$, our reduced grid size. We do this by Fourier transforming each bias field, nulling the modes higher than k_{Nyquist} , and inverse Fourier transforming back to real space. At k scales higher than this our hybrid bias model loses accuracy anyway and we don’t consider them in our inference, so we do not give up accuracy with this k -cut. We perform the same reduction on the velocity fields, cutting each component of the velocity vectors in

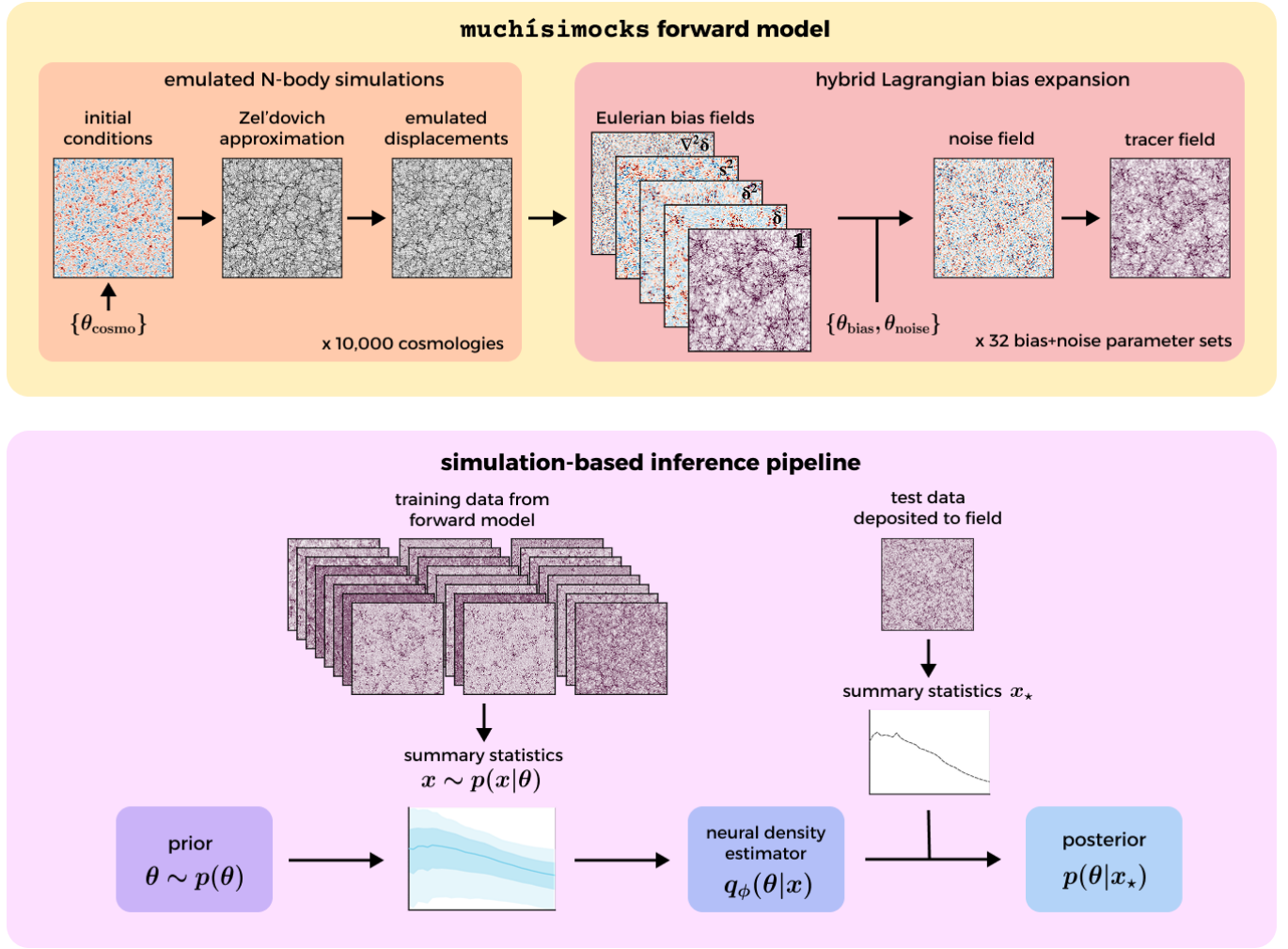


Figure 1. Schematic overview of the *muchisimocks* suite and simulation-based inference pipeline. For our forward model, we start with an initial conditions field based on input cosmological parameters, and apply the Zel’dovich approximation; these are passed to *map2map* which emulates the N -body displacement field. Using these as inputs to the hybrid Lagrangian bias expansion, we produce second-order Lagrangian bias fields advected to Eulerian space; these compose the core of the *muchisimocks* suite. These are then combined with a choice of bias parameters, as well as a full-field noise model. The combined tracer fields and associated parameters are used to train a neural posterior estimator, which conditioned on test data can then predict the associated parameters.

k -space. We finally perform a deconvolution of the bias fields, using a linear interpolation kernel at the original grid size N_{grid} to correct for the smoothing by the CIC kernel. (Technically the deconvolution should be performed before the k -cut, but the results are effectively equivalent and this ordering is much faster.) This deconvolution could be done at the summary statistics level, but we choose to deconvolve the fields for ease of use.

The final main suite, designed as a training set, contains 10,000 sets of tracer position fields (5 Lagrangian bias fields advected to Eulerian space) and velocity fields, each at a different cosmology; each is of size $(1h^{-1} \text{ Gpc})^3$ and resolution 128^3 . We also generate multiple smaller suites for testing, detailed in §3.5. We call these suites *muchisimocks* (a portmanteau of the words “muchísimos”, Spanish for “many”, and “mocks”).

2.3 Modeling noise at the field level

ksf: technically this is a postprocessing step only needed for the SBI part, where should it go?

We require a model for stochastic noise in order to model realistic discrete galaxy populations. This can be modeled at the level of each summary statistic, but our goal is to make this suite usable for a wide range of statistics as well as field-level analyses; we thus desire a field-level noise model. Further, as we are considering higher-order statistics, we need to include non-Gaussian noise contributions. We thus turn to the noise model of Rubira & Schmidt (2025), an EFT approach described by nonlinear couplings of a single Gaussian field to the bias fields. In this model the galaxy density field can be written as

$$\delta_g(\mathbf{x}) = \sum_{m=0}^{\infty} \sum_{\mathbf{O}} b_{\mathbf{O}}^{\{m\}} [\epsilon_G(\mathbf{x})]^m \mathbf{O}(\mathbf{x}) \quad (2)$$

where m is the order of the expansion in the noise field, $\epsilon_G(\mathbf{x}) = \mathcal{N}(0, 1)$ is a unit Gaussian field, $\mathbf{O}$ are operators of the galaxy density field, and $\{b_{\mathbf{O}}^{\{m\}}, b_{\mathbf{O}}^{\{m\}}\}$ are free coefficients. In our case we limit to second order in the bias expansion, as described in §2.2; limiting to zeroth order in m reduces to the noiseless case, as in Equation 1. For our SBI analysis, we include terms through first order in the noise

expansion, $m = 0, 1$. That gives us bias coefficients $\{b_O^{(0)}\} = b_1, b_2, b_{s^2}, b_\Delta$, as in Equation 1 (we fix $b_1^{(0)} = 1$), and noise coefficients $\{b_O^{(1)}\} = A_{\text{homog}}^{\text{noise}}, A_{b_1}^{\text{noise}}, A_{b_2}^{\text{noise}}, A_{b_{s^2}}^{\text{noise}}, A_{b_\Delta}^{\text{noise}}$.

We note that limiting the noise model to $m = 1$ means that we exclude noise contributions to $P_{gm}(k)$, as at $m = 1$ the noise field ϵ_G has zero mean (whereas $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ get factors of ϵ_G^2 which don't vanish). This means that the bias parameters may have to absorb partial noise contributions, though we show that we still achieve largely unbiased bias parameters; future work could extend this noise model to $m = 2$ or choose a well-motivated subset of noise parameters tailed to the statistics analyzed.

Operationally, we generate one unit Gaussian noise field ϵ_G per cosmology in our suite. We store these and use them to construct the noisy tracer field on-the-fly just before computing the summary statistics of the field. This means that we use the same noise realization when we select different bias and noise parameter sets for the same cosmology (detailed in §3.2 and §3.5), which introduces some undesirable non-independence; however, we expect the effects of this to be minimal, and the storage cost of one noise field per bias model is prohibitive. The prior ranges of the noise parameters used in our SBI analysis are shown in Table 1.

3 SIMULATION-BASED INFERENCE METHODOLOGY

3.1 Neural posterior estimation

The main idea behind simulation-based inference is that we can infer a posterior distribution of our data over the parameters of interest using only a *simulator*, which produces data samples given a set of parameters. This is distinct from explicit likelihood inference, in which we are required to define a likelihood of our data with respect to the parameters. Neural density estimators (NDEs) present a flexible and robust way to model a target distribution. Here, we use neural posterior estimation, which uses an NDE to directly model the posterior distribution.

To do this, we aim to minimize the distance between the true posterior $p(\theta|x)$ and the NDE model of the posterior $q_\phi(\theta|x)$, where ϕ are the parameters of the neural network model. We use maximum likelihood estimation, which minimizes the Kullback-Leibler divergence between the distributions. In practice, we consider the posteriors multiplied by $p(x)$:

ksf: update to using align

$$E = \dots \quad (3)$$

$$= L \quad (4)$$

where in the second line, we have converted the integral to a sum as we are using discrete samples from our simulation; in the third line, we observe that we are minimizing with respect to the neural network parameters ϕ and the true posterior $p(\theta|x)$ is independent of ϕ so it drops out; and in the third line we use the fact that minimizing the positive is the same as maximizing the negative.

Normalizing flows are a popular choice for the NDE, thanks to their expressivity and flexibility. The key idea is to transform an initial simple distribution into the target distribution through a chain of transformations. Specifically we use masked autoregressive flows (MAFs), which stack Masked Autoencoders for Distribution Estimation (MADEs) in order to model more complex distributions. This is defined as

$$q_\phi(\theta|x) = \dots \quad (5)$$

Table 1. Uniform prior ranges for the cosmological, bias, and noise parameters. The cosmological parameters are based on a ~ 10 range around the ? best-fit values, and the bias and noise parameters are loosely based on the hypervolume found by Zennaro et al. (2022) for a wide range of realistic galaxy populations.

Parameter	Lower bound	Upper bound
<i>Cosmological parameters</i>		
Ω_c	0.23	0.40
σ_8	0.65	0.90
h	0.6	0.8
Ω_b	0.04	0.06
n_s	0.92	1.01
<i>Bias parameters</i>		
b_1	−1.0	3.0
b_2	−2.0	2.0
b_{s^2}	−2.0	2.0
b_Δ	−10.0	10.0
<i>Noise parameters</i>		
$A_{\text{homog}}^{\text{noise}}$	−3.0	3.0
$A_{b_1}^{\text{noise}}$	−3.0	3.0
$A_{b_2}^{\text{noise}}$	−2.0	2.0
$A_{b_{s^2}}^{\text{noise}}$	−5.0	5.0
$A_{b_\Delta}^{\text{noise}}$	−10.0	10.0

The MAF samples an initial base variable z_0 from a Gaussian $\mathcal{N}$, and then performs the transformations $z_t = f_t(z_{t-1}, \theta)$, where f_t is neural network that takes as input the previous variable z_t as well as the parameters θ . The Jacobian (the determinant term) arises from the change of variables, and involves f^{-1} , thus requiring f to be invertible. The invertibility property of NFs both enables us to sample from the network and gives us an explicit density, which is key for performing SBI. ksf: via tucci paper, check The loss function derived from the KL divergence (Equation 4) is used to train the weights and biases of the network.

3.2 Training set

To train our simulation-based inference model, we use the main muchisimocks suite of bias fields to generate a set of tracer fields. Our prior on the cosmological parameters is set by the range of the suite, as described in 2.1. We need to define the range of the bias coefficients $\{b\}$ we use to sum the fields; for this, we use the results of Zennaro et al. (2022), which define priors that capture a range of realistic galaxy populations modeled with the extended subhalo abundance matching technique ?. We choose to use uniform priors with slightly wider ranges than the relations found by Zennaro et al. (2022), a more conservative choice; these prior ranges are shown in Table 1. For the noise coefficients $\{A^{\text{noise}}\}$ we choose wide uniform priors loosely based off of the corresponding (noiseless) bias coefficients.

While we are limited to the 10,000 cosmologies defined in the muchisimocks suite, we can combine these into tracer fields with any number of bias parameter sets we choose. At a given cosmology, we then have multiple tracer fields generated with distinct sets of bias and noise parameters. Though this means the final tracer fields are not fully independent of each other, this has been shown (e.g. in Hahn et al. (2023c)) to improve inference by significantly augmenting the training data size and allowing the model to disentangle cosmology and bias parameters. Note that the summing of bias fields with the

chosen bias coefficients is essentially instantaneous and can be done in postprocessing just before computing summary statistics, and thus requires no additional compute time or storage.

As we later want to test convergence of the inference with the number of bias parameter sets, we use a nested latin hypercube to generate the bias parameters. The top-level LH contains 320,000 points, resulting in 32 independent bias parameter sets for each of the 10,000 cosmologies. Nested subsets of this LH are also LHs, with 16, 8, 4, 2, and 1 bias parameter sets per cosmology. This was achieved with the Qian (2009) algorithm for nested latin hypercubes, which works by . . . *ksf: TODO*. A python implementation of this algorithm along with validation tests are available in the `gethypercube`¹ package. *ksf: should i include an appendix detailing the nested-LH package? or just link to the github as im doing? or does it deserve it's own little software paper?* This is shown to achieve LHs at each level that have properties much closer to a true LH than either taking random subsets of a top-level LH, or stacking bottom-level LHs. The use of LH-distributed parameters has been shown to be important for accurate and robust training, so we expect that this nested strategy leads to robustness in our convergence tests, which are a key result of this work. We generate one nested LH for the noiseless case with the 4-parameter set of bias parameters; and a separate nested LH for the noisy case with 4 bias parameters and 5 noise parameters, for a total of 9 independently varied parameters.

To generate our final tracer fields, we multiply the *muchisimocks* bias fields and noise field by the selected set of corresponding bias and noise parameters as in Equation (2), and sum these weighted fields. We normalize these fields by the original `map2map` grid size (512^3) before computing their summary statistics.

3.3 Summary statistics

We perform our SBI analysis with our *muchisimocks* framework on three standard summary statistics: the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter power spectrum $P_{gm}(k)$, and the bispectrum $B_{ggg}(k_1, k_2, k_3)$.

The galaxy–galaxy auto power spectrum $P_{gg}(k)$ is defined by

$$\langle \delta_g(\mathbf{k}) \delta_g(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} + \mathbf{k}') P_{gg}(k), \quad (6)$$

where $k = |\mathbf{k}|$ is the Fourier space wavenumber. This is the two-point function of the galaxy field, the most common statistic for cosmological inference from galaxy samples that captures all the Gaussian information. We use 32 logarithmically spaced bins from $k = 0.01$ to $k = 0.4$; the maximum k value is set by the Nyquist frequency of our mocks. We use the BACCO code to compute $P_{gg}(k)$ for our mocks. We can compute the 15 cross-terms of the 5 bias fields and later sum their contributions based on the bias parameters to obtain the power spectrum of the total tracer field; in practice we do this for the noiseless case, but in the noisy case compute $P_{gg}(k)$ for the total tracer field as this would add additional terms.

The galaxy–matter cross power spectrum $P_{gm}(k)$ is defined by

$$\langle \delta_g(\mathbf{k}) \delta_m(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} + \mathbf{k}') P_{gm}(k), \quad (7)$$

where $\delta_m(\mathbf{k})$ is the matter density field in Fourier space. This represents the galaxy–galaxy lensing contribution to 3x2pt analyses, and is critical for constraining galaxy bias and breaking parameter degeneracies. We use the same binning as $P_{gg}(k)$, and again use the BACCO code to compute $P_{gm}(k)$ for our mocks; for our fiducial results, we only use up to $k_{\max} = 0.25$ for $P_{gm}(k)$.

The galaxy auto-bispectrum $B_{ggg}(k_1, k_2, k_3)$ is defined by

$$\langle \delta_g(\mathbf{k}_1) \delta_g(\mathbf{k}_2) \delta_g(\mathbf{k}_3) \rangle = (2\pi)^3 \delta_D(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) B_{ggg}(k_1, k_2, k_3). \quad (8)$$

This is the Fourier space three-point function. We choose 12 linearly spaced bin edges from $k = 0.01$ to $k = 0.4$, yielding 203 triangles; for our fiducial results, we only use triangles with edges up to $k_{\max} = 0.25$, yielding 50 triangles. We use the `PolyBin3D` code (?) to compute $B_{ggg}(k_1, k_2, k_3)$ for our mocks.

The summary statistics computed on the *muchisimocks* training set are shown in Fig. 2.

3.4 Implementation and training

Before training, we reparameterize the cosmology and bias parameters to combinations that are better constrained by the summary statistics, and thus should result in improved training convergence (?). *ksf: ask Bea for citation* We multiply the bias and noise parameters by the relevant power of σ_8 , obtaining $\sigma_8 b_1$, $\sigma_8^2 b_2$, $\sigma_8^2 b_{s^2}$, $\sigma_8 b_\Delta$, $\sigma_8 A_{b_1}^{\text{noise}}$, $\sigma_8^2 A_{b_2}^{\text{noise}}$, $\sigma_8^2 A_{b_{s^2}}^{\text{noise}}$, and $\sigma_8 A_{b_\Delta}^{\text{noise}}$. When sampling from the posterior, we can divide out the appropriate power of the recovered *sig* to obtain estimates of the original bias and noise parameters. We also do not include the cosmological parameters h , Ω_b , and n_s in the SBI training, as we do not expect to constrain these parameters (and confirmed in tests); these are still varied in the training set and implicitly marginalized over.

We use the publicly available `sbi` ? package for the NPE and MAF implementations (??); the MAF uses the `nflows` Python package (??). Our data vector is comprised of the binned statistics discussed in §3.3, concatenated when using multiple statics. The statistics combinations we show are: $P_{gg}(k)$, $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$, $P_{gg}(k) + P_{gm}(k)$, $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ (we exclude $B_{ggg}(k_1, k_2, k_3)$ and $P_{gm}(k)$ alone, as these do not contain much constraining power on their own and in practice are almost always considered in combination with $P_{gg}(k)$). We use a 90%/10% training/validation split of our training set. We use the `ADAM` optimizer, and stop training when the loss over the validation loss does not improve for 20 epochs; we save the network weights at the best validation loss. We perform a sweep over several hyperparameters, using 30 random samples from the following sets: the learning rate $\{10^{-2}, 3 \times 10^{-3}, 10^{-3}, 3 \times 10^{-4}, 10^{-4}, 3 \times 10^{-5}, 10^{-5}\}$, hidden features $\{32, 64, 128, 256\}$, training batch size $\{32, 64, 128, 256\}$, and number of transforms $\{4, 6, 8, 10\}$. The sweeps are performed separately for each statistic or combination of statistics, and for each choice of noiseless (bias parameters only) and noisy (bias and noise parameters); we always use the maximum size training set of 320,000 data vectors, and for the statistics use $k_{\max}^{P_{gg}} = 0.4$, $k_{\max}^{P_{gm}} = 0.25$, and $k_{\max}^B = 0.25$. The model which achieves the minimum best validation loss is selected; we use this best set of hyperparameters to train variations of the model, such as when varying the number of training points and the $k_{\max}$ values. The best hyperparameters for the fiducial $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ noisy case are: learning rate 10^{-4} , hidden feature vector of size 256, batch size 256, and 6 transforms. The set of best hyperparameters vary across the statistics, with the noisy case (which varies more parameters) generally preferring larger hidden feature vectors and batch sizes, and most learning rates around 10^{-4} .

Training on the maximum size training set of 320,000 data vectors takes a few hours on a single CPU for the fiducial model described above. To display the posteriors and compute covariances and other statistics from them, we draw 10,000 samples from the posterior; we

¹ <http://github.com/kstoreyf/gethypercube>

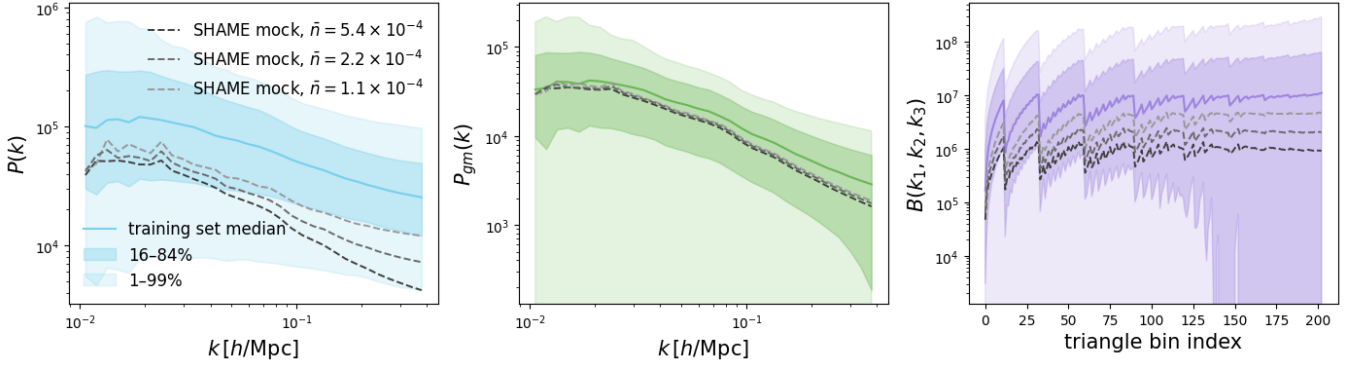


Figure 2. The summary statistics computed on the *muchisimocks* training set: the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter power spectrum $P_{gm}(k)$, and the galaxy bispectrum $B_{ggg}(k_1, k_2, k_3)$. The median, 16th–84th percentiles, and 5th–95th percentiles of the training set. The out-of-distribution test mock catalog constructed with extended subhalo abundance matching (SHAME) on a BACCO simulation is also shown (black dashed). In our fiducial inference model, scales below $k_{\max} = 0.25$ for $P_{gm}(k)$ and triangles with sides smaller than $k_{\max} = 0.25$ for $B_{ggg}(k_1, k_2, k_3)$ are excluded.

confirm that the posteriors are generally converged with this number of samples.

3.5 Test sets

We generate several test sets to validate our *muchisimocks* suite and our SBI approach.

Our first test set is a *cosmic variance set*. For this we use *muchisimocks* pipeline to construct 1000 additional mocks each with a different random seeds, all at the same fixed cosmology; we choose this to be the same cosmology as the out-of-distribution SHAME mock test described below. We fix the bias parameters to those measured from that mock, and use these to combine the bias fields into a tracer field for each realization. For the noise parameters, we cannot measure these from the mock directly; to identify realistic noise parameters, we generate 1000 sets of noise parameters and add the resulting noise field to each of the 1000 mocks, obtaining tracer fields that have the same cosmology and bias parameters but different noise parameters. We compute the statistics of these fields, and then compute the reduced χ^2 between them and the statistics of the SHAME mock. Specifically, we use $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ with the fiducial $k_{\max}$ choices (the noise model does not affect $P_{gm}(k)$ as discussed in §2.3 so we exclude it), and choose the set of noise parameters that gives the lowest χ^2 . We then use this set of noise parameters and regenerate the noise fields with different noise realizations, and combine this with the biased tracer fields to obtain our final set of 1000 cosmic variance mocks at fixed cosmology, bias, and noise parameters, only varying the random seeds. For the recovery tests, we compute the mean statistic over these 1000 mocks, and perform SBI on that mean to reduce the randomness from cosmic variance; in Appendix A we show that this gives the same result as combining the results of inference on each of the 1000 mocks, while being much faster.

Our second test set is our *coverage set*. This consists of 1000 mocks generated with the *muchisimocks* pipeline that span the same prior range of cosmology, bias, and noise parameters as in the training set. To select the parameters, we construct an independent 1000-point LH for the cosmologies, and a separate LH for the bias and noise parameters. The coverage set tests the performance of our method across the prior space and its dependence on the parameter values.

Our final test set is our *SHAME mock*; we construct a realistic

catalog of galaxies using the extended subhalo abundance matching (SHAME) approach ? applied to a BACCO N -body simulation ?. This is a stringent test as it uses both a different, significantly higher resolution underlying cosmological simulation and a completely different galaxy formation model than the *muchisimocks* data used to train the inference network, testing the generalizability of our model. In detail, we use a simulation with a box size of $(1024 h^{-1} \text{Mpc})^3$ (slightly larger than the *muchisimocks* training data) and 3072³ particles, which uses the fixed-and-paired technique (?), and has Planck cosmology $\Omega_c = 0.309$, $\sigma_8 = 0.816$, $h = 0.677$, $\Omega_b = 0.0486$, and $n_s = 0.967$ (?). Halos are identified using a variant of the SUBFIND halo finder (?), and we populate these halos with galaxies using the SHAME model with the following parameter choices: $\sigma_{\log M} = 0.47513517$, $t_{\text{merger}} = -0.4577029$, $f_{k, \text{cen+sat}} = 0.20147321$, $f_{k, \text{cen-sat}} = -0.02964998$, $\beta_{\text{lum}} = 14.82600407$. We choose a fiducial number density of $\bar{n} = 2.2 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$, similar to that of DESI luminous red galaxies (LRGs) when scaled to $z = 0$. ksf: checking that this is right? also are the shame parameters chosen to be lrg-like? To check our inference results, we measure the bias parameters of this sample using the field-level probabilistic bias approach of ?; we find $b_1 = 0.474$, $b_2 = 0.127$, and $b_{s^2} = -0.339$ (we do not expect b_{Δ} to be well-measured by this approach). We also test a higher and lower number density sample, of $5.4 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$ and $1.1 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$; these are measured to have slightly different bias parameters. The summary statistics computed on the SHAME mock are shown in Fig. 2; they largely fall within the middle 68% of the training set, with the exception of $P_{gg}(k)$ at smaller scales, which reaches the edge of the training set distribution but still falls within it.

For each of the two simulations of opposite phases from the fixed-and-paired pair, the SHAME model gives a set of discrete galaxy tracers. We deposit these to a mesh using the Cloud-in-Cell mass assignment scheme, of size $(512)^3$. We then remove the high- k modes and deconvolve the field in the same way as the *muchisimocks* fields. We compute the statistics on each of the two fields with the same method used for the training data, and take the mean of these two statistics as the data vector for inference.

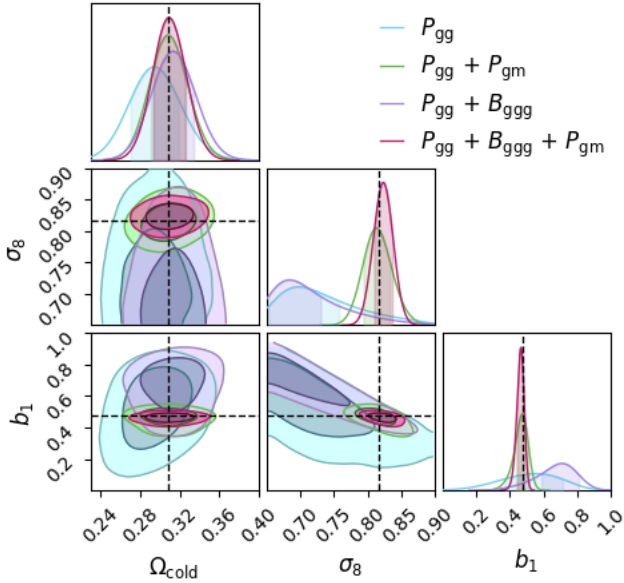


Figure 3. Posteriors for the key parameters Ω_m , σ_8 , and b_1 inferred from the mean of the statistics over our cosmic variance test set, for each of the statistics combinations considered. The inclusion of $P_{gm}(k)$ breaks the $\sigma_8 - b_1$ degeneracy, providing tight and unbiased constraints on the parameters; adding $B_{ggg}(k_1, k_2, k_3)$ tightens the constraints. The other bias parameters and the noise parameters are inferred and marginalized over; the full posterior plot is shown in Fig. B1.

4 SBI RESULTS

4.1 In-distribution recovery tests

4.1.1 Cosmic variance test set results

We first validate our SBI pipeline on the cosmic variance test set, at fixed values of the cosmology, bias, and noise parameters. We compute summary statistics on the test data in the same way as on the training data.

For the cosmic variance test set, we compute the mean of all of the statistics computed on the 1000 individual realizations and use this as the data vector for inference. In Appendix A, we show that this recovers the typical size of posterior of the individual samples (and errors wider than the distribution of the means of the individual realizations). This allows us to visualize a typical contour, and average out the effects of cosmic variance to visualize the mean constraint. The posteriors are shown in Figure 3 for three key parameters, Ω_m , σ_8 , and b_1 ; we infer and marginalize over the other bias parameters and the noise parameters. (The full contour plot is shown in Fig. B1.)

We find that we obtain tight, unbiased constraints on the cosmology and bias parameters when using the full combination of statistics ($P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$), validating the inference approach.

While $P_{gg}(k)$ on its own provides significant constraining power on Ω_m , it results in a weak constraint on b_1 and a weak, biased constraint on σ_8 . The weakness of these constraints is expected as a result of the perfect degeneracy between σ_8 and b_1 in the power spectrum (as $P_{gg}(k) = (b_1 \sigma_8)^2 \hat{P}(k)$); we check that the combination $\sigma_8 b_1$ is accurately constrained. Including $B_{ggg}(k_1, k_2, k_3)$ interestingly only slightly increases the precision on Ω_m ; further, it does not improve the bias on σ_8 . Notably, it does significantly increase the precision on the higher-order biases parameters b_2 and b_{s2} . We

see that using the statistics combination $P_{gg}(k) + P_{gm}(k)$ produces unbiased constraints on σ_8 and b_1 ; this makes sense as $P_{gm}(k)$ is able to break this degeneracy (as $P_{gm}(k) = b_1 \sigma_8^2 \hat{P}(k)$). This corroborates our hypothesis that the bias observed in σ_8 for $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ is likely a result of projection effects caused by a lack of constraining power. The full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ preserves the unbiased constraints, and the bispectrum adds an additional improvement of constraining power compared to $P_{gg}(k) + P_{gm}(k)$: it improves Ω_m by XX, σ_8 by XX, and b_1 by XX, and notably the bias parameters b_2 by XX, b_{s2} by XX, and b_Δ by XX. *ksf: TODO figure out which data to use to quantify this*

We note that these results are for real space, and we would expect differences in the relative information content in the statistics in redshift-space, where there is additional cosmological information contained in redshift-space distortions.

4.1.2 Coverage test set results

We infer the parameters of the 1000 samples of our coverage test set, spanning the prior range of our training set; this allows us to test the dependence of the estimated parameters on the true parameter values. Note that the prior has a wide range of values of the noise parameters, so these test samples will have a variety of noise levels.

In Figure 4, we show the differences between the predicted values and true values of the parameters as a function of the true value, which spans the full training prior; we display the median of the 1000 mocks, and the 16-84th percentile range. The predicted value is the mean of the inferred posterior; we check that the behavior is very similar for the maximum a posteriori (MAP) values. For $P_{gg}(k)$, we can see that the range of residuals is relatively wide, especially for b_1 , and for all key parameters show a stronger bias with larger distance from the center of the prior. This is expected when the statistic does not provide significant constraining power on the parameter, as is the case with $P_{gg}(k)$, as the prediction will tend towards the mean. Including $B_{ggg}(k_1, k_2, k_3)$ improves the strength of the parameter-dependent bias for all parameters, and also generally tightens the range of the residuals. When we consider $P_{gg}(k) + P_{gm}(k)$, we can see that the bias on σ_8 improves significantly, as $P_{gm}(k)$ breaks the degeneracy with b_1 as we saw in the cosmic variance test set. $P_{gm}(k)$ also improves the constraints on Ω_m and b_1 . Further including $B_{ggg}(k_1, k_2, k_3)$ helps marginally, and this full parameter combination yields largely unbiased results across the coverage set, indicating that our inference is robust across the prior. There are still slight biases for $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ particularly at high Ω_m and σ_8 , so the analysis is most reliable towards the center of the prior.

The residuals shown above only quantify the accuracy of the means of the posteriors; to assess the inferred uncertainties, we compute the coverage of the samples, defined as the fraction of samples for which the true value falls within the predicted uncertainty range, as a function of the given level of uncertainty. For instance, for a credibility level of 68%, 68% of the samples should have the truth fall within their predicted 68% error region (“empirical coverage”). A higher empirical coverage compared to credibility level indicates underconfidence, while lower indicates overconfidence. This coverage (or probability-probability) plot is shown in Figure 5, for the 500 mocks of the coverage set closer to the center of the prior in Ω_m , σ_8 , and Ω_m . We see that the inferred uncertainties are generally quite well calibrated for all three key parameters. Using only $P_{gg}(k)$ or $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$ results in a slight underconfidence in

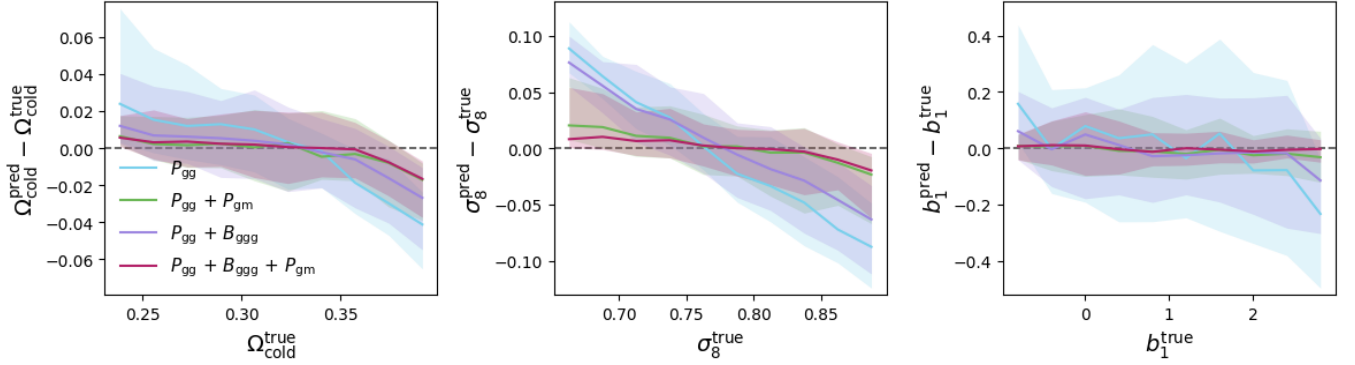


Figure 4. The difference between the predicted (mean of the posterior) and true parameter values for the key parameters for each of the 1000 coverage test set mocks, as a function of the true value. The median and 16th-84th percentile ranges in bins of the true value are shown for each of the statistics combinations considered. Using the full statistics combination gives largely unbiased recovery across the prior range, with a small biases at the high ends of Ω_m and σ_8 ; using fewer statistics results in stronger parameter-dependent biases that tend towards the mean. Including more statistics also results in tighter parameter constraints, most notably with $P_{gm}(k)$ significantly improving the precision and bias on all key parameters.

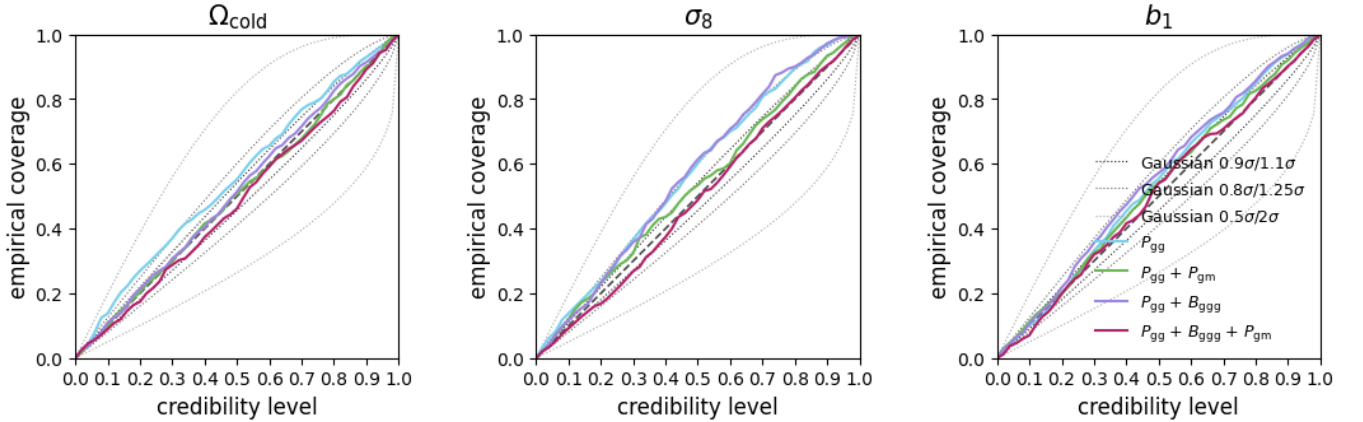


Figure 5. Coverage (probability–probability) plots for the key parameters for the 500 mocks of the coverage test set closest to the center of the prior. The empirical coverage is the fraction of test set points with predicted uncertainties at the given credibility level that encompass the true value. The closeness to the 1:1 line indicates well-calibrated uncertainties; dotted lines show the expected deviations if posterior widths were off by the given factor of σ , assuming Gaussians centered on the truth. The posteriors are largely well-calibrated, with the full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ giving the best calibration across parameters.

the posteriors, at the level of $0.1 - 0.25\sigma$ (assuming a Gaussian distribution centered on the truth). The full statistics combination $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ results in improved calibration of the uncertainties, with only slight fluctuations. We note that for this check, we have excluded the mocks on the edges of the prior to isolate the assessment of the SBI-inferred uncertainties; when we make the coverage plot for all 1000 mocks of the coverage set, we do see some overconfidence. This is a result of the biases towards the edges of the prior which cause the uncertainty regions to not sufficiently cover the truth, and the uncertainty estimates do not fully compensate for this bias.

These results show that our inference pipeline can accurately recover the cosmological and bias parameters across a range of parameter values.

4.2 Out-of-distribution recovery test: SHAMe mock

The results of our SBI pipeline on our SHAMe out-of-distribution test mock with fiducial number density $\bar{n} = 2.2 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$ are shown in Figure 6 for the key parameters (the full contours are shown in Figure B2). We obtain generally unbiased constraints on the cosmological and bias parameters when using the full statistics combination, with Ω_m , σ_8 , and b_1 all within 1σ from the true values. The trends for the statistics generally follow those seen in the cosmic variance test set (§4.1 and Figure 3), with $P_{gm}(k)$ breaking the degeneracy between σ_8 and b_1 , and $B_{ggg}(k_1, k_2, k_3)$ improving constraints significantly for the bias parameters and moderately for the cosmological parameters. For the SHAMe data, we obtain a XXX% improvement in precision on Ω_m and XXX% on σ_8 when including $P_{gm}(k)$ compared to $P_{gg}(k)$ alone; additionally including $B_{ggg}(k_1, k_2, k_3)$ results in XXX% and XXX% improvements on Ω_m and σ_8 respectively compared to $P_{gg}(k) + P_{gm}(k)$.

We do see slight biases in the values of b_1 and b_2 when using the

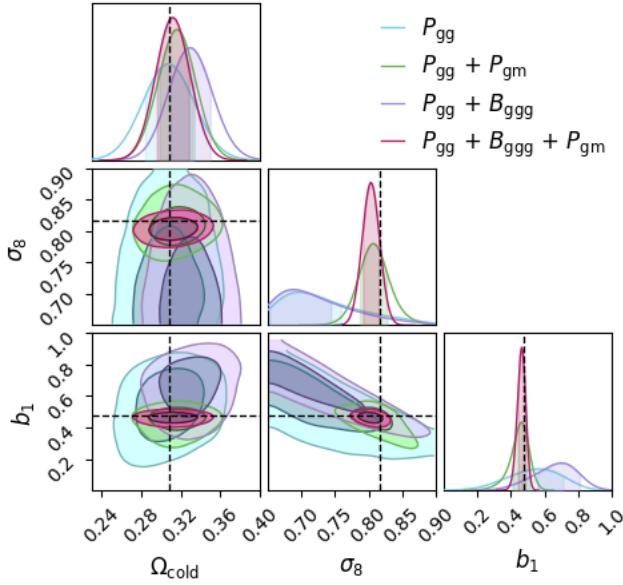


Figure 6. Posteriors for the key parameters Ω_m , σ_8 , and b_1 for our out-of-distribution SHAMe mock test, for each of the statistics combinations considered. The recovered parameters with the full statistics combination are unbiased within 1σ , and the statistics show similar constraining power as in the in-distribution test. The full posterior plot is shown in Fig. B2.

full statistics combination, at the $XXX\sigma$ and $XXX\sigma$ levels, while these were on average accurately constrained in the cosmic variance test set. This is likely a result of the bias parameters absorbing some model mismatch, which could be an issue for approaches such as imposing tighter priors on bias parameters; however, for our inference the cosmological parameters remain unbiased.

We show the results for the full parameter combination on the SHAMe samples with a range of number densities in Figure 7. We find that the constraints on the cosmological parameters are robust to the noise level across this range. The constraints on the bias parameter b_1 do lose precision at lower number densities; this could be related to degeneracies with the noise parameters. We note that the noise parameters themselves are poorly constrained, as seen in the full contour plot (Figure B2). However, we check that the noise model is necessary for properly inferring the parameters; assuming noiseless training data results in strong biases in the inference.

This test on out-of-distribution data demonstrates the robustness of our forward model to modeling realistic galaxy populations constructed with completely different models.

4.3 Scale dependence

We investigate the dependence of our results on the maximum k scale included in our inference, for each of the statistics. We keep the maximum scale for $P_{gg}(k)$ fixed to $k_{\max} = 0.4 \, h\text{Mpc}^{-1}$, the Nyquist frequency of the box. At this scale, the map2map emulator of N -body displacements is shown to achieve significantly sub-percent accuracy ($\sim 0.2\%$) on the power spectrum (Jamieson et al. 2023), and Pellejero Ibanez et al. (2023) showed that map2map combined with the hybrid Lagrangian bias expansion to second order achieves 1-2% precision to $k_{\max} = 0.6 \, h\text{Mpc}^{-1}$. (Previous works using the hybrid bias expansion with a power spectrum emulator find accuracy down to $k_{\max} = 0.6 \, h\text{Mpc}^{-1}$ (Hadzhiyska et al. 2021; ?).) For the

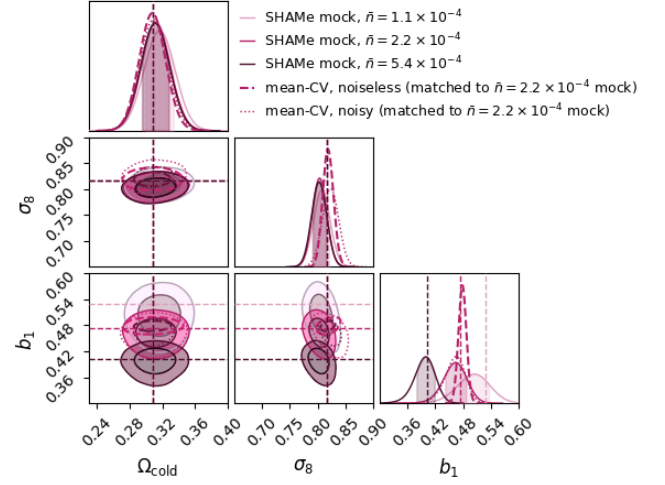


Figure 7. Impact of noise on the inferred posteriors, for the mean of $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ over the cosmic variance test set. The brown contours show inference trained on statistics computed on noiseless realizations, and inferred with a model with no noise; the pink contours show inference computed on realizations injected with the Rubira & Schmidt (2025) noise model, and inferred with the same noise model. Even with perfect knowledge of the noise model, the constraints on σ_8 and b_1 degrade significantly.

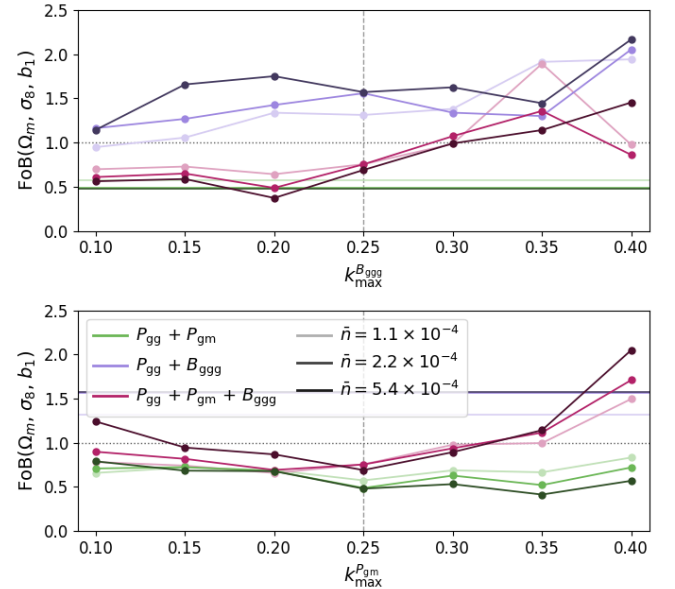


Figure 8. Scale dependence of the figure of bias (FoB) for the three key parameters combined for the SHAMe out-of-distribution mock. The top panel shows the FoB as a function of $k_{\max}$ for the bispectrum, fixing to our fiducial $k_{\max} = 0.25$ for $P_{gm}(k)$; the bottom panel shows it as a function of $k_{\max}$ for $P_{gm}(k)$, fixing to our fiducial $k_{\max} = 0.25$ for the bispectrum. In both cases, the FoB for the full statistics combination remains around or below 1σ until $k_{\max} = 0.25$, after which the inference becomes biased.

bispectrum, ? recently validated the hybrid bias expansion down to $k_{\max} = 0.25 \, h\text{Mpc}^{-1}$. ksf: anything to cite/mention for the cross-power-spectrum?

In Figure 8, we show the figure of bias for the SHAMe out-of-distribution mock for the three key parameters Ω_m , σ_8 , and b_1 combined, as a function of the $k_{\max}$ of $B_{ggg}(k_1, k_2, k_3)$ and $P_{gm}(k)$; when varying $k_{\max}^{B_{ggg}}$, we fix $k_{\max}^{P_{gm}} = 0.25 \, h\text{Mpc}^{-1}$, and when varying $k_{\max}^{P_{gm}}$, we fix $k_{\max}^{B_{ggg}} = 0.25 \, h\text{Mpc}^{-1}$. We see that using only $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$ results in a bias moderately above 1σ for all number density samples, regardless of $k_{\max}$; this is likely largely a result of projection effects. When including the cross spectrum, $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$, the inference is accurate to 1σ up until $k_{\max}^{B_{ggg}} = 0.25 \, h\text{Mpc}^{-1}$, aligning with the result validating the hybrid bias expansion on the bispectrum and indicating that the limiting factor is the expansion rather than the emulated displacements. The statistics combination $P_{gg}(k) + P_{gm}(k)$ results in accurate inference for all number densities through $k_{\max}^{P_{gm}} = 0.4 \, h\text{Mpc}^{-1}$. However, when additionally adding $B_{ggg}(k_1, k_2, k_3)$, the inference becomes biased pushing below $k_{\max}^{P_{gm}} = 0.25 \, h\text{Mpc}^{-1}$. We also check the full grid of combinations, and find $k_{\max}^{B_{ggg}} = 0.25 \, h\text{Mpc}^{-1}$, $k_{\max}^{P_{gm}} = 0.25 \, h\text{Mpc}^{-1}$ to be the optimal combination of limiting scales; pushing either or both to smaller scales results in further bias. It is initially odd that pushing $P_{gm}(k)$ to smaller scales results in a bias only when including the $B_{ggg}(k_1, k_2, k_3)$; we speculate that this could be because our noise model does not account for noise in $P_{gm}(k)$, though further investigation is merited.

4.4 Dependence on number of cosmologies and bias parameter sets

We explore the convergence of our inference with the number of training data points. We vary the number of cosmologies by selecting only a subset of cosmologies to train the network with, choosing $N_{\text{cosmo}} = 500, 1000, 2000, 4000, 6000, 8000, 10000$; the subsets are nested to minimize random variance. To vary the number of bias parameters, we train on each layer of the nested Latin hypercube of bias and noise parameters, resulting in a range of bias parameters per cosmology: $N_{\text{bias}} = 1, 2, 4, 8, 16, 32$. For each of these combinations of the number of cosmology and bias parameters, we train a new model using the fiducial hyperparameters for the given statistics combination, and perform inference on the 1000 mocks of the coverage set.

Figure 9 shows the figure of merit and figure of bias for the three key parameters combined, averaged over the coverage set, for the full statistics combination as a function of the total number of training points $N_{\text{cosmo}} \times N_{\text{bias}}$, shown as separate lines for each choice of N_{cosmo} . We see that the FoM increases with increasing number of training points, steeply until $N_{\text{train}} = 10,000$ and then more moderately. The FoM does not plateau even at 320,000 training points, indicating that the inference is still not converged and the precision would continue to improve with additional training data. The FoB similarly reduces sharply until $N_{\text{train}} = 10,000$ where it crosses $\text{FoB} = 1\sigma$, but then more-or-less plateaus, indicating that above a few tens of thousands of training points, the training data is sufficient for unbiased inference.

We also find that there is not a strong trend with the number of training cosmologies, as shown by the lines for the different cosmologies largely overlapping. With a smaller number of training data points (below $N_{\text{train}} = 20,000$), it is more important to have more cosmologies rather than more bias parameter sets, shown by the spread in

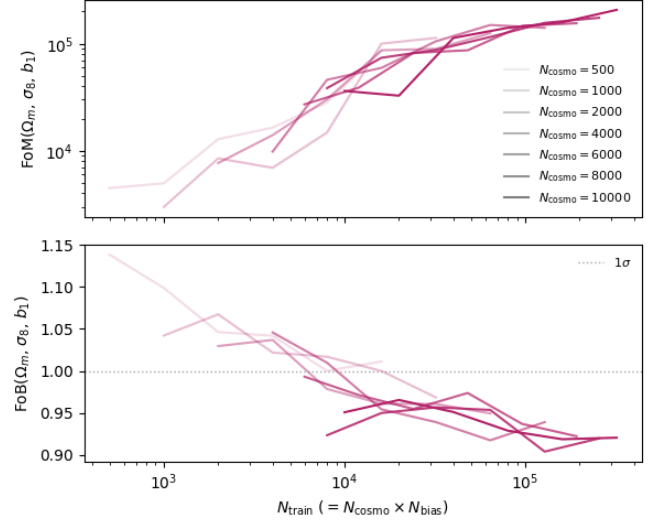


Figure 9. Convergence of the inference with training set size, for $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ on the coverage test set for the three key parameters Ω_m , σ_8 , and b_1 combined. The top panel shows the figure of merit (FoM), and the bottom panel shows the figure of bias (FoB), normalized by the 68th percentile of the χ^2 distribution for three dimensions (corresponding to the three parameters); the median values over the coverage set are shown. The opacity indicates the number of training cosmologies. Increasing the number of training points increases the precision significantly until $N_{\text{train}} \sim 20,000$ and then more marginally, with the number of cosmologies and bias parameter sets contributing similar precision; the bias also plateaus around $N_{\text{train}} \sim 20,000$, with the number of cosmologies mattering more than the number of bias parameters for smaller N_{train} . ksf: make higher res

the FoB around that value; however, beyond $N_{\text{cosmo}} = 2000$, it is sufficient to add more bias parameter sets per cosmology. This somewhat surprising finding is encouraging given that generating training data at varying cosmologies is more expensive than populating each cosmology with varying bias parameter sets. Previous SBI analyses already use around this number of training cosmologies with full N -body codes (without the need for emulators, e.g. the Quijote latin hypercube); our result shows that this may be sufficient in terms of cosmologies, but that additional constraining power and robustness could be gained by populating these with many more galaxy formation parameter sets per cosmology.

In Figure 10, we show the FoM and FoB for all the statistics combinations, for each key parameter individually. Note that for a Gaussian sample, we would expect $\text{FoB} \approx 0.8$; our FoB stays slightly higher than this for all the parameters, bottoming out at 0.85–1.0, possibly as a result of non-Gaussian behavior. We see that for σ_8 , a small number of training cosmologies already saturates the precision for $P_{gg}(k)$ and $P_{gg}(k) + \text{bispec}$, though around $N_{\text{train}} = 20,000$ are needed to reach a plateau in the FoB below 1σ . Similarly for b_1 for those statistics combinations, the precision begins to plateau around the same number of training points, and seems to saturate around $N_{\text{train}} = 100,000$. The statistics combinations $P_{gg}(k) + P_{gm}(k)$ and $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ show steep improvements in both precision and accuracy until $N_{\text{train}} = 10,000$ for all parameters, and then improve more moderately. Notably, by our maximum number of training points $N_{\text{train}} = 320,000$, the FoM has not yet plateaued, indicating that additional training data may continue to improve the inference and we have not yet extracted the full information content of the statistics. We also again find that for all parameters, increasing the

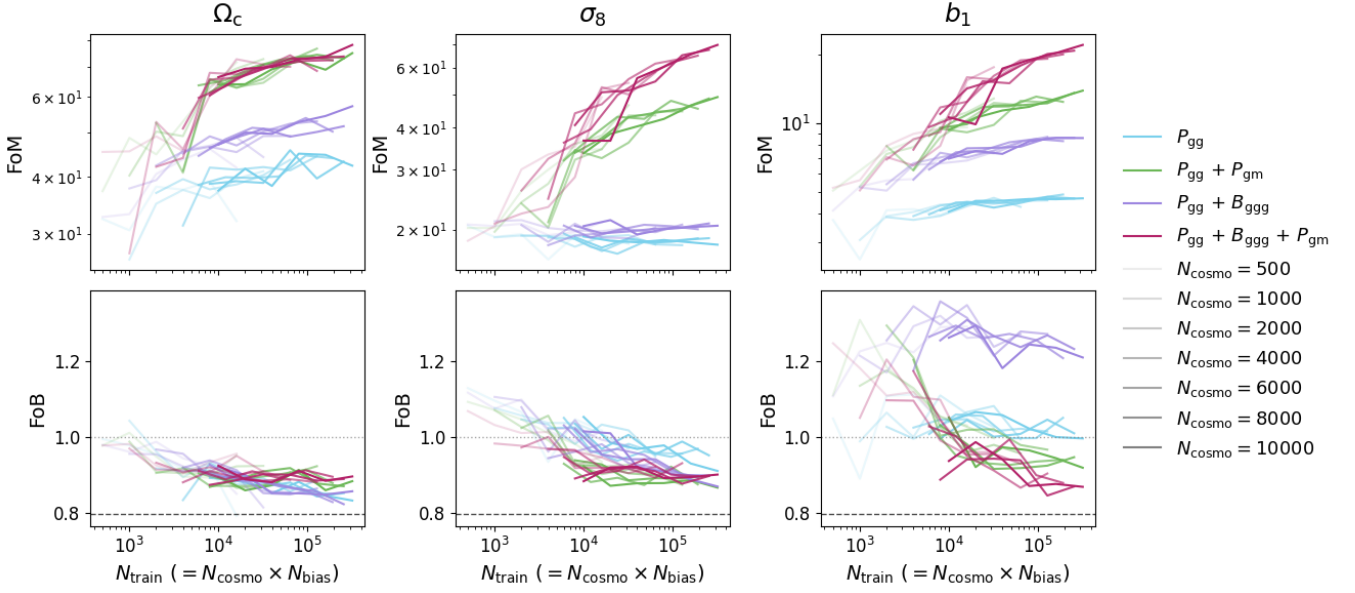


Figure 10. Convergence of the inference with training set size, for all the statistics combinations on the coverage test set for the three key parameters Ω_m , σ_8 , and b_1 individually. The top row shows the figure of merit (FoM) and the bottom row shows the figure of bias (FoB); the median values over the coverage set are shown. The opacity indicates the number of training cosmologies, and the dashed line indicates the expectation for an unbiased Gaussian distribution. The FoM increases with training set size for most statistics and parameters, with the exception of σ_8 with $P_{gg}(k)$ and $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$ which do not show improvement; all have an FoB around or below 1σ .

number of bias parameter sets per cosmology results in similar FoM and FoB values as increasing the number of cosmologies, beyond a few thousand cosmologies.

5 DISCUSSION

There are many possible extensions to the model. We could extend the bias model to the third order, which has been shown to be important when... We could even generate larger volumes using map2map; volume is a main limitation of current SBI approaches.

There are many further potential uses of the *muchisimocks* suite. For instance, one could learn to reconstruct the velocity fields given the density fields. The suite could be used to explore the information in higher-order statistics of the density field, including optimal neural network compressions. One could also investigate the effect of more restrictive priors on the Lagrangian bias parameters, encoding our knowledge specific galaxy samples.

The smallest scales we can include are limited by... *ksf: when limited by map2map vs HBE?*

6 SUMMARY AND CONCLUSIONS

ACKNOWLEDGEMENTS

The authors would like to thank Beatriz Tucci, Ivana Nikolac, Matt Ho, and Jed Homer; Matteo Zennaro, Daniel López-Cano, Francisco Maion and the rest of the BACCO group; and Risa Wechsler and the KIPAC Galaxy Formation and Cosmology group for helpful discussions.

SOFTWARE AND LLM USAGE

The following software were used in this work: *ksf: fill in software*. In this work, large language models (LLMs) were used to make code modifications, write basic code such as plotting functions, support background research, and provide basic L^AT_EX and citation support. Core code and all writing did not use LLMs.

DATA AVAILABILITY

The *muchisimocks* suite and the code used in this paper will be made publicly available at [thisurl](#).

REFERENCES

- Doeser L., Jamieson D., Stopyra S., Lavaux G., Leclercq F., Jasche J., 2024, *Monthly Notices of the Royal Astronomical Society*, 535, 1258
- Hadzhiyska B., García-García C., Alonso D., Nicola A., Slosar A., 2021, *Journal of Cosmology and Astroparticle Physics*, 2021, 020
- Hahn C., Beutler F., Sinha M., Berlind A., Ho S., Hogg D. W., 2018, Likelihood Non-Gaussianity in Large-Scale Structure Analyses, <https://arxiv.org/abs/1803.06348v1>, doi:10.1093/mnras/stz558
- Hahn C., et al., 2022, Λ CDM S_{IM}BIG: A Forward Modeling Approach To Analyzing Galaxy Clustering (arXiv:2211.00723), doi:10.48550/arXiv.2211.00723
- Hahn C., et al., 2023a, Λ CDM S_{IM}BIG: The First Cosmological Constraints from the Non-Linear Galaxy Bispectrum (arXiv:2310.15243), doi:10.48550/arXiv.2310.15243
- Hahn C., et al., 2023b, Λ CDM S_{IM}BIG: The First Cosmological Constraints from Non-Gaussian and Non-Linear Galaxy Clustering (arXiv:2310.15246), doi:10.48550/arXiv.2310.15246
- Hahn C., et al., 2023c, *Journal of Cosmology and Astroparticle Physics*, 2023, 010
- Jamieson D., Li Y., de Oliveira R. A., Villaescusa-Navarro F., Ho S., Spergel D. N., 2023, *The Astrophysical Journal*, 952, 145
- Jamieson D., Li Y., Villaescusa-Navarro F., Ho S., Spergel D. N., 2024, Field-Level Emulation of Cosmic Structure Formation with Cosmology and Redshift Dependence (arXiv:2408.07699), doi:10.48550/arXiv.2408.07699
- Lemos P., et al., 2023, SimBIG: Field-level Simulation-Based Inference of Galaxy Clustering (arXiv:2310.15256), doi:10.48550/arXiv.2310.15256
- Massara E., et al., 2024, {sc SimBIG}: Cosmological Constraints Using Simulation-Based Inference of Galaxy Clustering with Marked Power Spectra (arXiv:2404.04228), doi:10.48550/arXiv.2404.04228
- Nguyen N.-M., Schmidt F., Tucci B., Reinecke M., Kostić A., 2024, *Physical Review Letters*, 133, 221006
- Pellejero-Ibanez M., Stuecker J., Angulo R. E., Zennaro M., Contreras S., Arico G., 2022, *Monthly Notices of the Royal Astronomical Society*, 514, 3993
- Pellejero Ibanez M., Angulo R. E., Jamieson D., Li Y., 2023, Hybrid-Bias and Displacement Emulators for Field-Level Modelling of Galaxy Clustering in Real and Redshift Space (arXiv:2307.09134)
- Qian P. Z. G., 2009, *Biometrika*, 96, 957
- Rubira H., Schmidt F., 2025, Non-Gaussian Galaxy Stochasticity and the Noise-Field Formulation (arXiv:2511.05484), doi:10.48550/arXiv.2511.05484
- Tucci B., Schmidt F., 2024, *Journal of Cosmology and Astroparticle Physics*, 2024, 063
- Zennaro M., Angulo R. E., Contreras S., Pellejero-Ibáñez M., Maion F., 2022, *Monthly Notices of the Royal Astronomical Society*, 514, 5443
- de Santi N. S. M., et al., 2023, *The Astrophysical Journal*, 952, 69

APPENDIX A: IMPACT OF VARIANCE AND NUMBER DENSITY

We investigate the behavior of the posteriors of the cosmic variance test set, which contains 1000 realizations run with different seeds at the same cosmology, bias, and noise parameters. Our goal is to be able to quantify and visualize the accuracy and precision of the inference on this test set.

We first compute the statistics on all 1000 realizations, and perform the inference on each individually with our SBI pipeline. We obtain a posterior for each of the 1000 realizations. We then look at the distribution of the means of these posteriors. To visualize the error regions, we compute the mean-of-the-means of the posteriors, and shift each posterior such that it is centered on that value; we then pool the posteriors, drawing 500 samples from each. This shows the typical size and shape of the error region, centered on the mean.

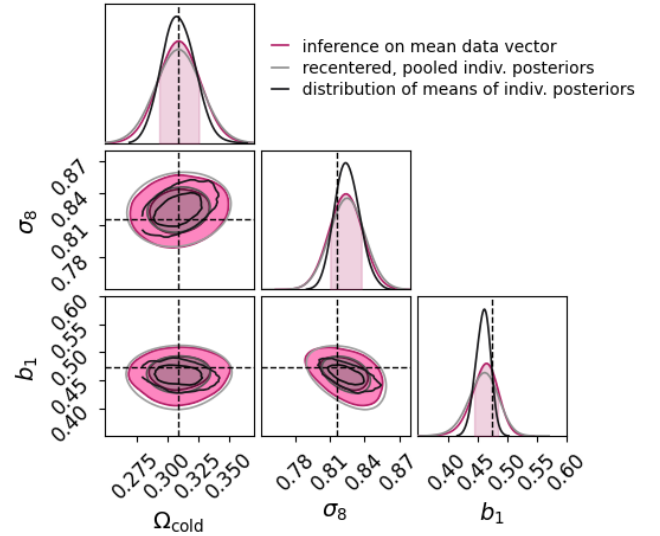


Figure A1. Validation of inference on the mean of the statistics for the cosmic variance test set, for the full $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ combination. The grey contours show 10 posteriors of inference on statistics randomly selected realizations from the test set, and the pink contours show inference on the mean of 1000 statistics over all the realizations. The shape and width of the posteriors are consistent, with the mean falling in the middle of the individual posteriors. ksf: bc network is focusing on low dim representation ksf: in each projection, get peak of dist (or mean); and plot 2d hist of the grey means. and the 1ds, will be a hist of grey vs the contours ksf: dont need to focus on uncertainties ksf: importance sampling of the posteriors of individuals - interpolate to get posterior value for each point in param space. then could combine. can share data with marcos ksf: test with fewer than 1000 to see if an issue with number of points

We also compute the mean of the statistics of the 1000 mocks, and perform the inference on that mean data vector. This mean data vector has the benefit of reducing the effect of cosmic variance on the statistic, but brings the complication of having a different covariance than the data the SBI model was trained on (namely, much lower variance). In standard MCMC approaches, this would be handled by appropriately scaling the covariance matrix, but in the context of SBI, the covariance is learned by the model from the training data and is not given explicitly. Thus it is important to understand the behavior of the model when we use the mean data vector for inference.

Figure A1 shows the result of these three quantities on the cosmic variance test set, all for the combined statistic $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$. The black contour shows the distribution of the means of the 1000 posteriors; the grey contour shows the recentered, pooled posteriors; and the magenta contour shows the inference performed on the mean of the 1000 data vectors. We see that the mean-of-the-means inference lies nearly exactly on top of the recentered, pooled inferences. This indicates that despite the lower variance of the mean data vector, the SBI pipeline produces a posterior that has the typical width of a posterior estimated with the noisier data vector, as that was the noise level of data it was trained on. We thus use the inference on the mean data vector throughout the paper to be able to show the typical result of the inference, with the effect of cosmic variance (which would cause a bias in any given individual sample) removed.

We also see that the distribution of the means of the individual realizations is significantly tighter than the width of a typical individual posterior, especially for b_1 . This indicates that our inference is

underconfident, predicting generally broader posteriors than would a perfectly calibrated model. This should be taken into account when interpreting the results of this analysis. However, the mean of distribution of means agrees with the true values of the parameters, as well as with the inference on the mean, indicating that our results are unbiased.

ksf: i am not quite sure how to square the results of these two plots. when we perform inference on the mean of 1000, we get the same posterior size as any individual. but when we use the higher number density shame model, we do get a smaller posterior size compared to the lower number density one. (i think the noiseless one makes a bit more sense, as that's actually a different model where we also train on noiseless data. but the different noise levels of shame are all using the same model with the same training data.)

APPENDIX B: FULL POSTERIOR CONTOURS

This paper has been typeset from a $\text{\TeX}/\text{\LaTeX}$ file prepared by the author.

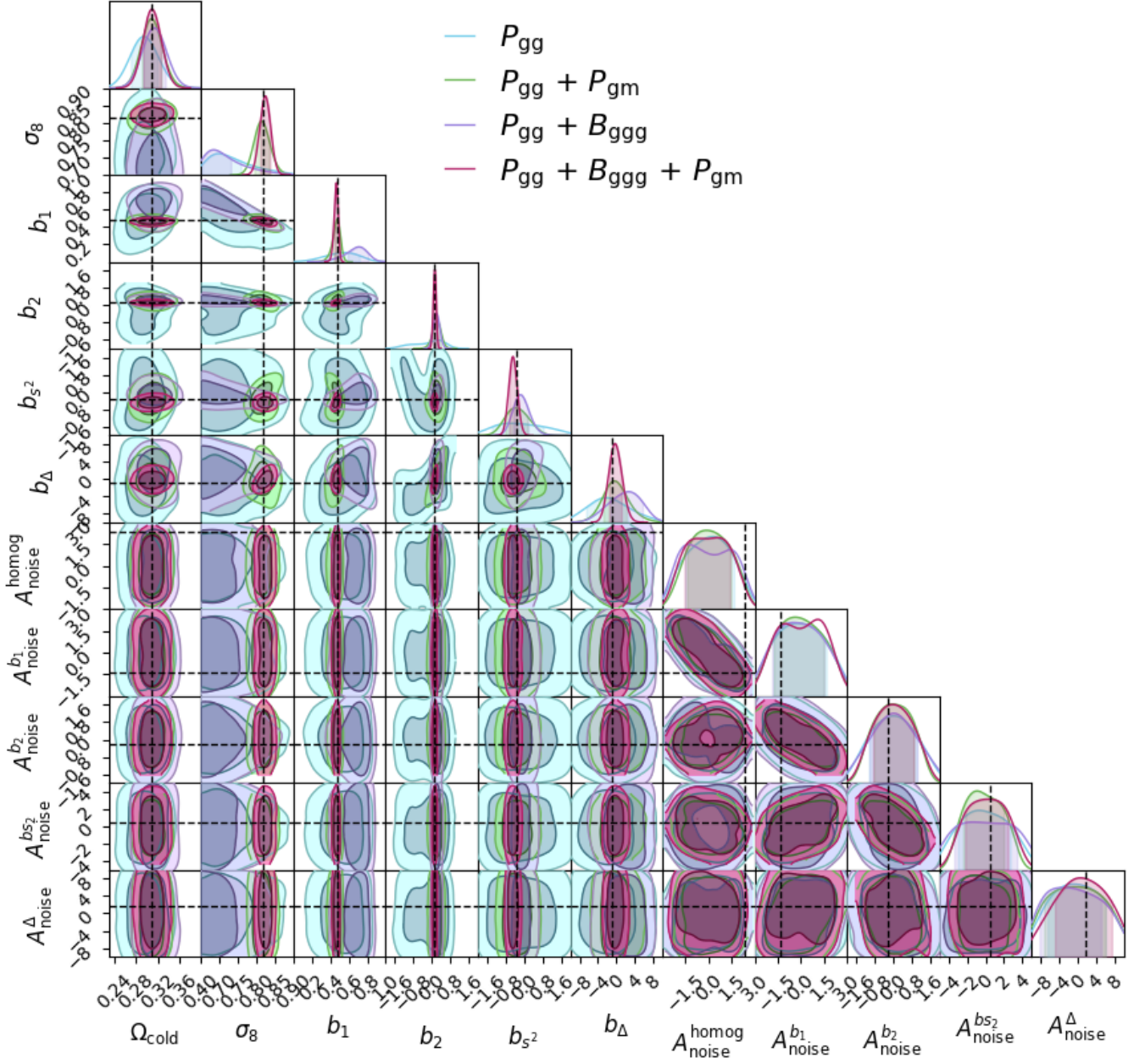


Figure B1. Full posterior contours for the cosmic variance test set.

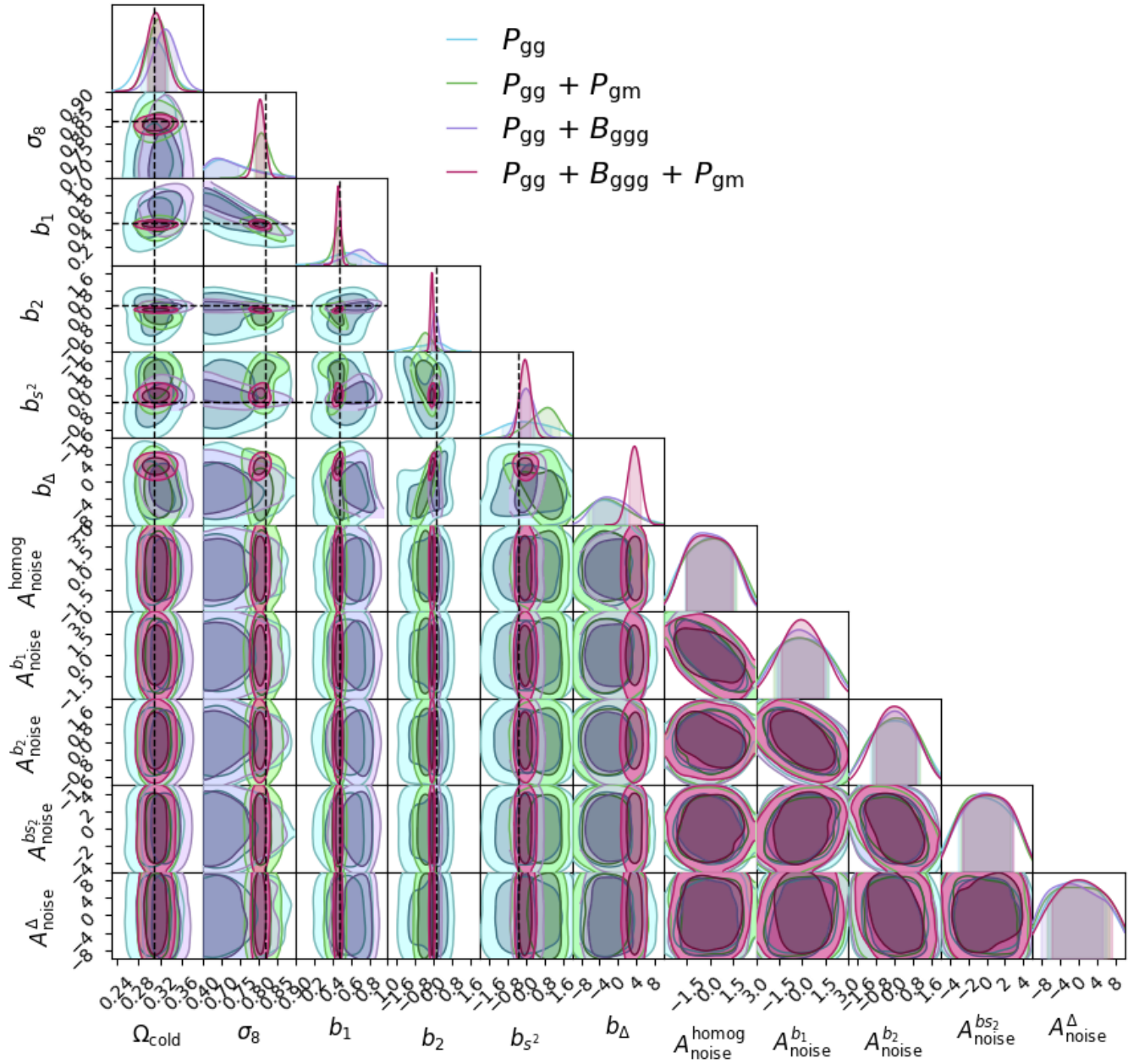


Figure B2. Full posterior contours for the SHAME test set.